고급유저를 위한 LabVIEW 프로그래밍 Tips & Tricks

한국NI 교육센터
전문강사 박 진 호

MAINTENANCE 64%

INTEGRATION 8%

UNIT TESTING 7%

DESIGN 6%

CODING 5% SPECIFICATIONS 5% REQUIREMENTS 5%



MOVE



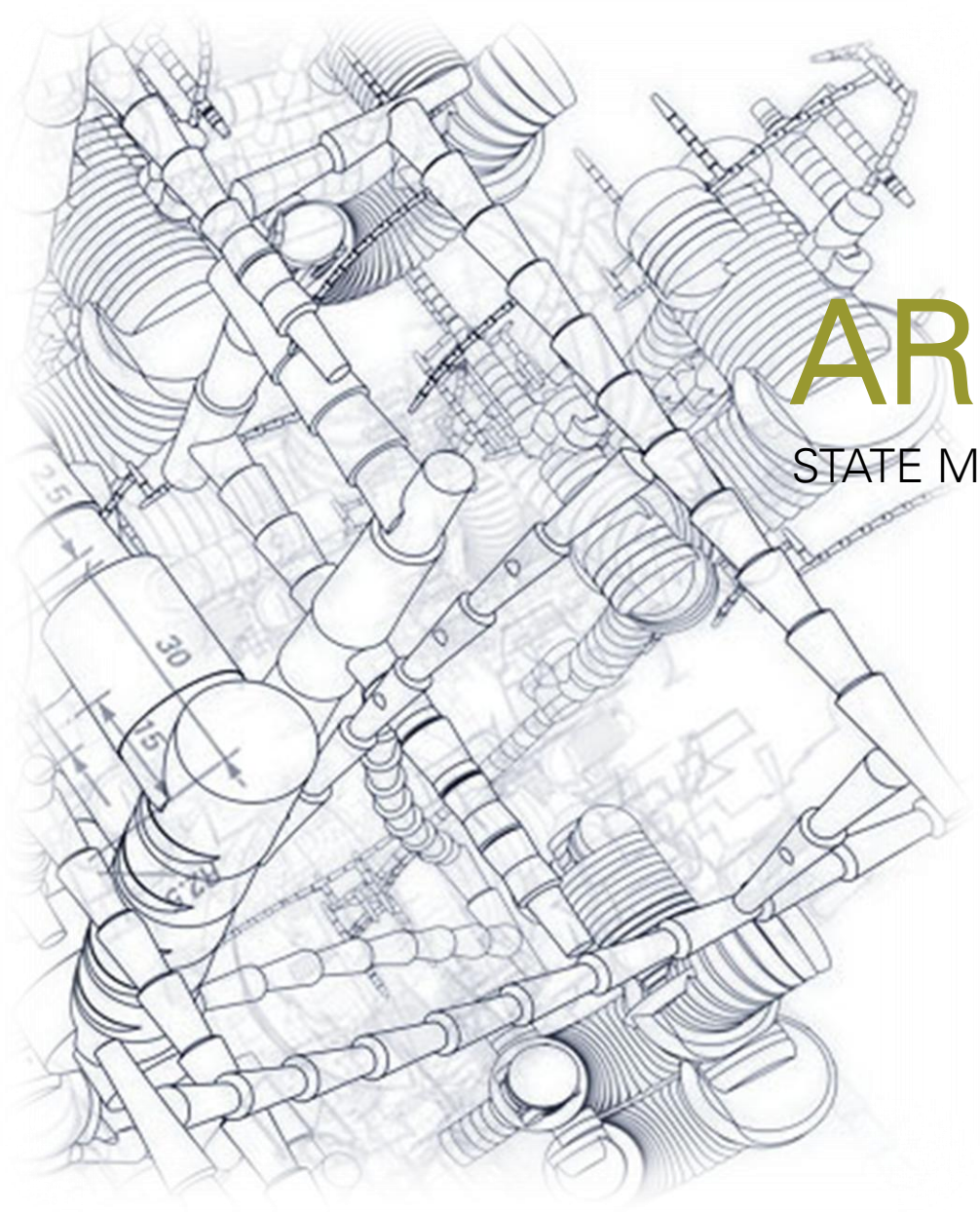
EAT



SLEEP

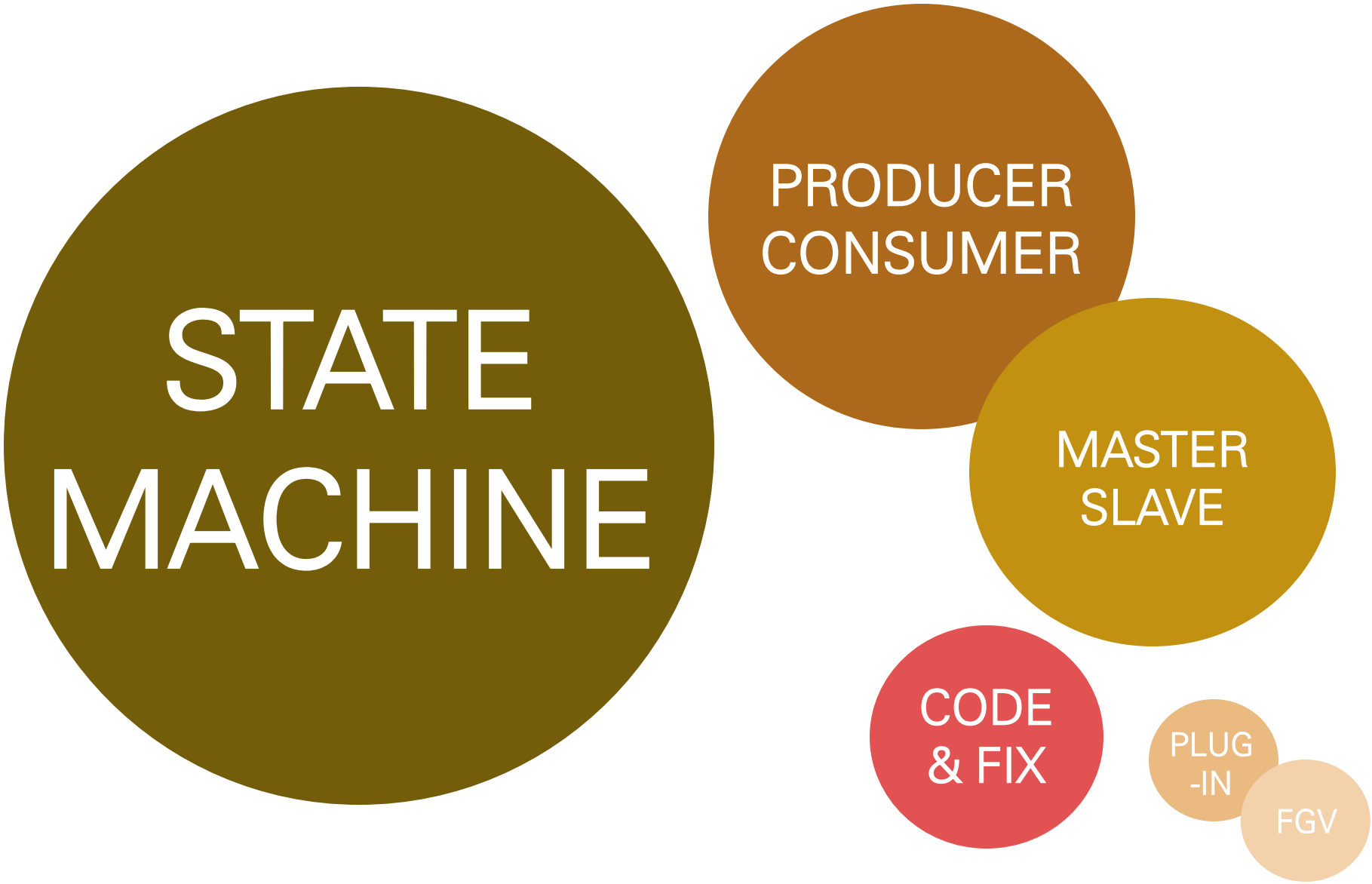
AND...


CODING!!
CODING!!
CODING!!

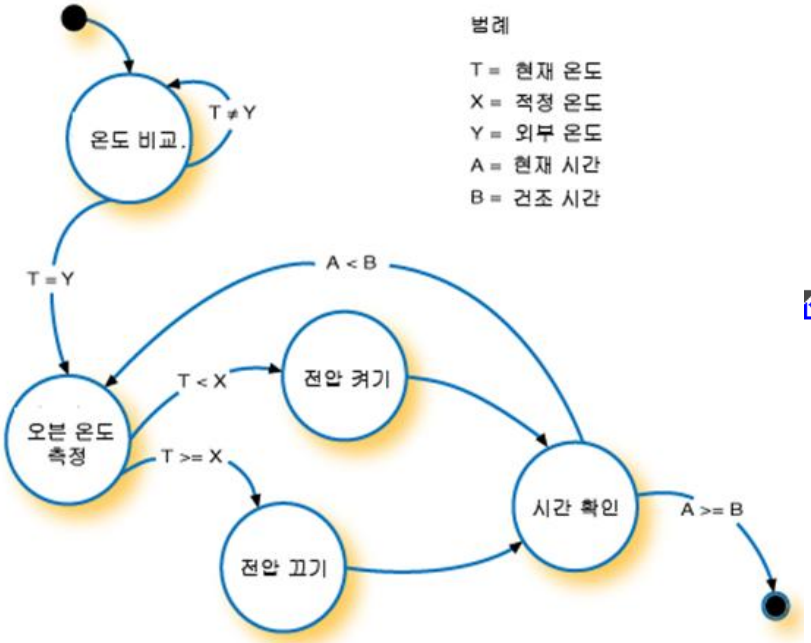


ARCHITECTURE

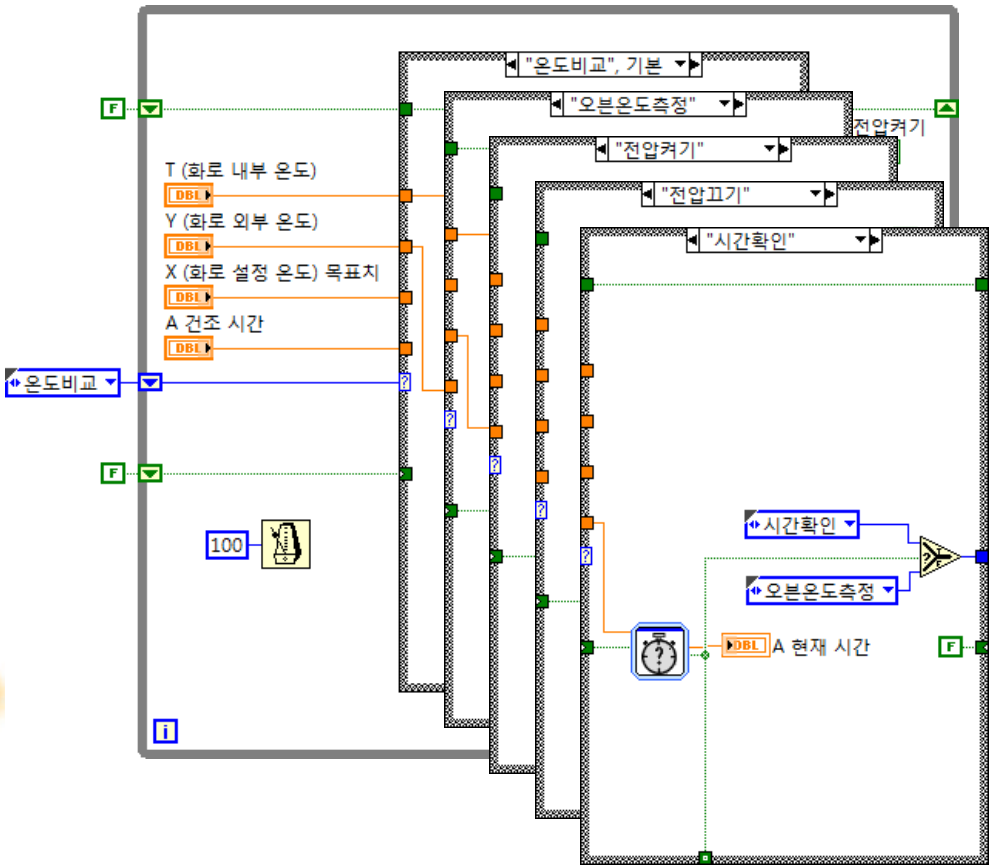
STATE MACHINE / PLUG-IN / FGV


STATE MACHINE
PRODUCER CONSUMER
MASTER SLAVE
CODE & FIX
PLUG -IN
FGV

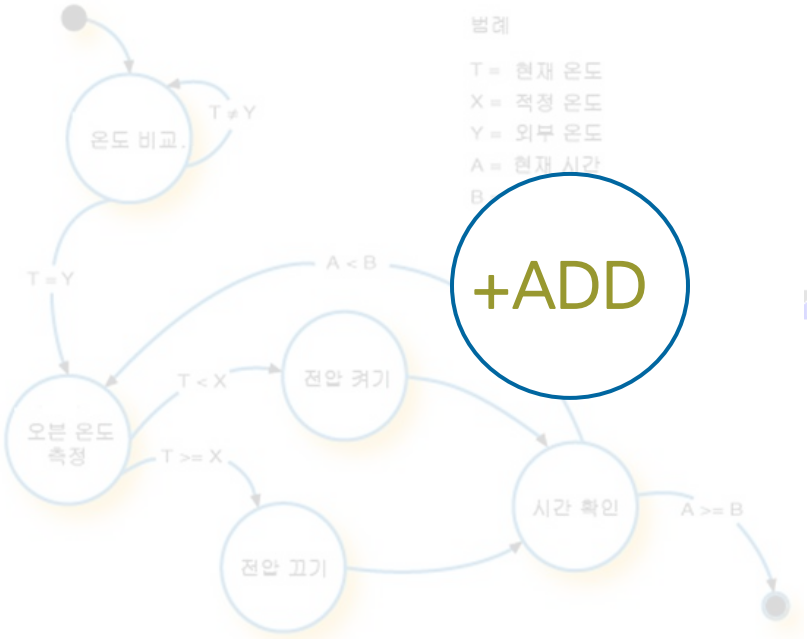
STATE DIAGRAM



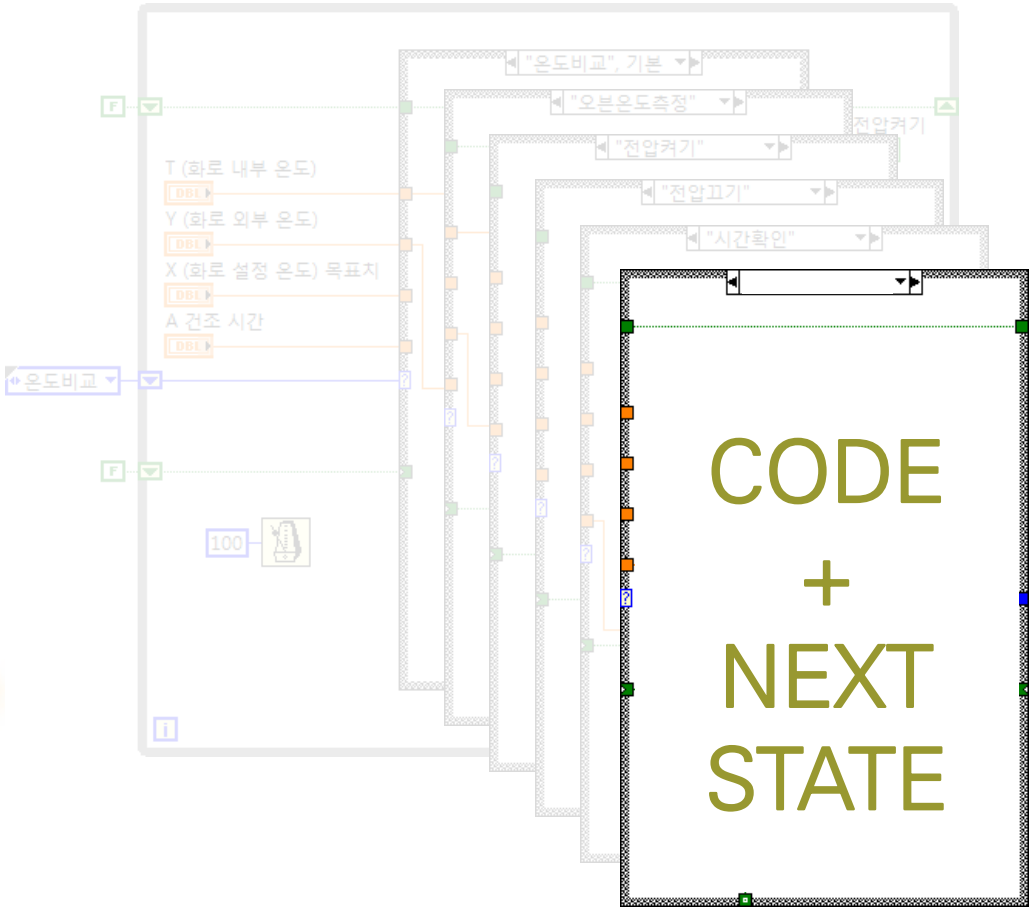
STATE MACHINE



STATE DIAGRAM



STATE MACHINE

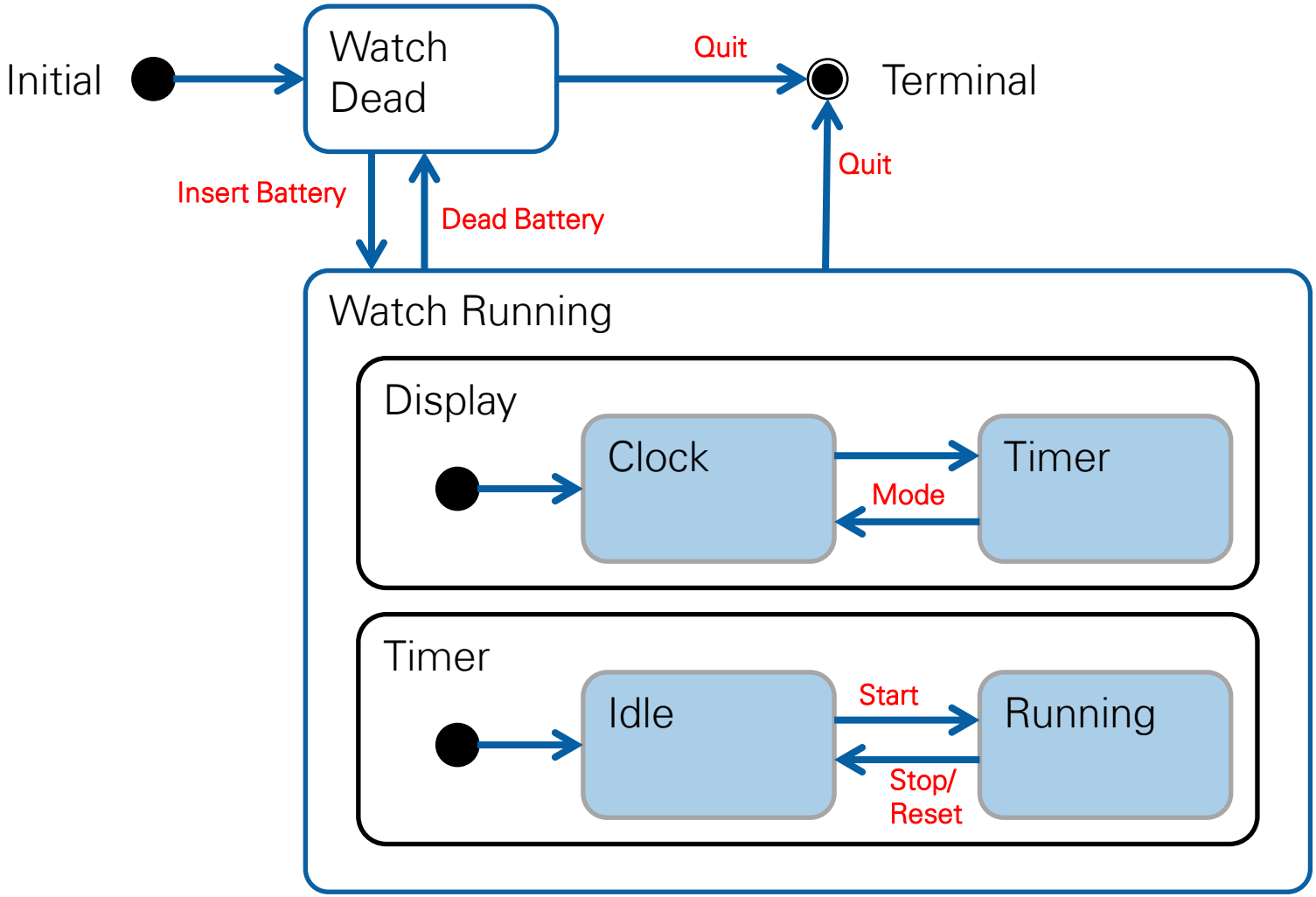


COMMUNICATION

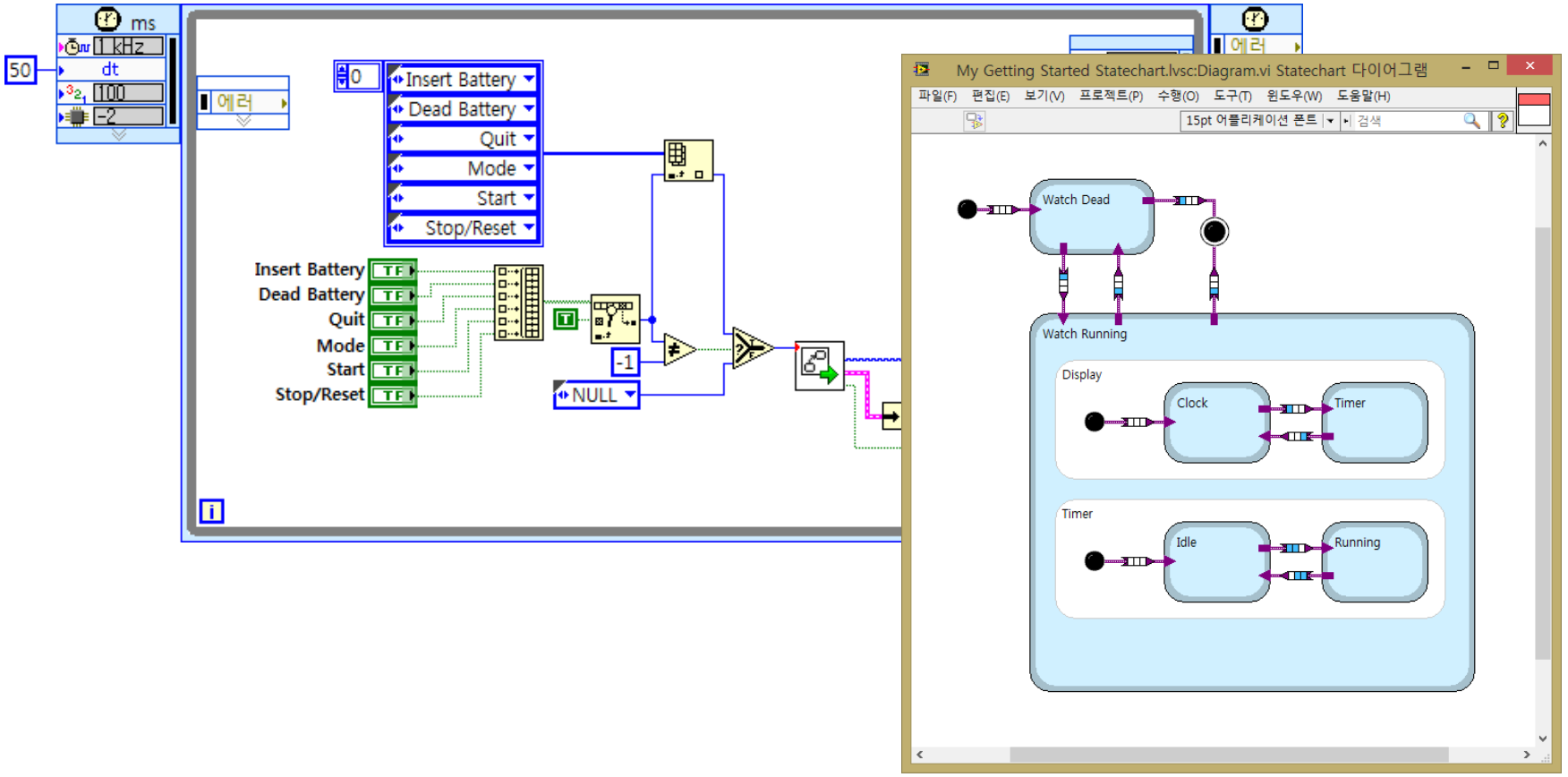
시뮬레이션 전자시계 프로그램 개발을 의뢰합니다.
전자시계는 크게 **현재시간**과 **스톱워치**의 두 가지 기능이 있습니다.
먼저 Insert Bettery를 눌러 전원을 인가해야 모든 작동이 시작됩니다.
작동 중 Dead Battery를 누르면 모든 작동 멈춤니다.
또한 Quit 버튼을 누르면 어떤 상태이건 프로그램을 종료합니다.
외부에 Mode, Start, Stop/Reset의 세 버튼이 존재하며 시간 디스플레이 창이 있습니다.
전원을 인가했을때 항상 현재시간이 먼저 표시됩니다.
Mode를 누르면 현재시간과 스톱워치가 번갈아 가며 표시됩니다.
만일 스톱워치가 동작하는 중 Mode를 눌러도 스톱워치는 동작해야 합니다.
스톱위치는 Start 버튼을 눌러 시작하며 Stop/Reset 버튼을 누르면 정지합니다. 정지 상태에서 Stop/Reset 버튼을 한번더 누르면 시간은 0으로 리셋됩니다... ...



STATE DIAGRAM

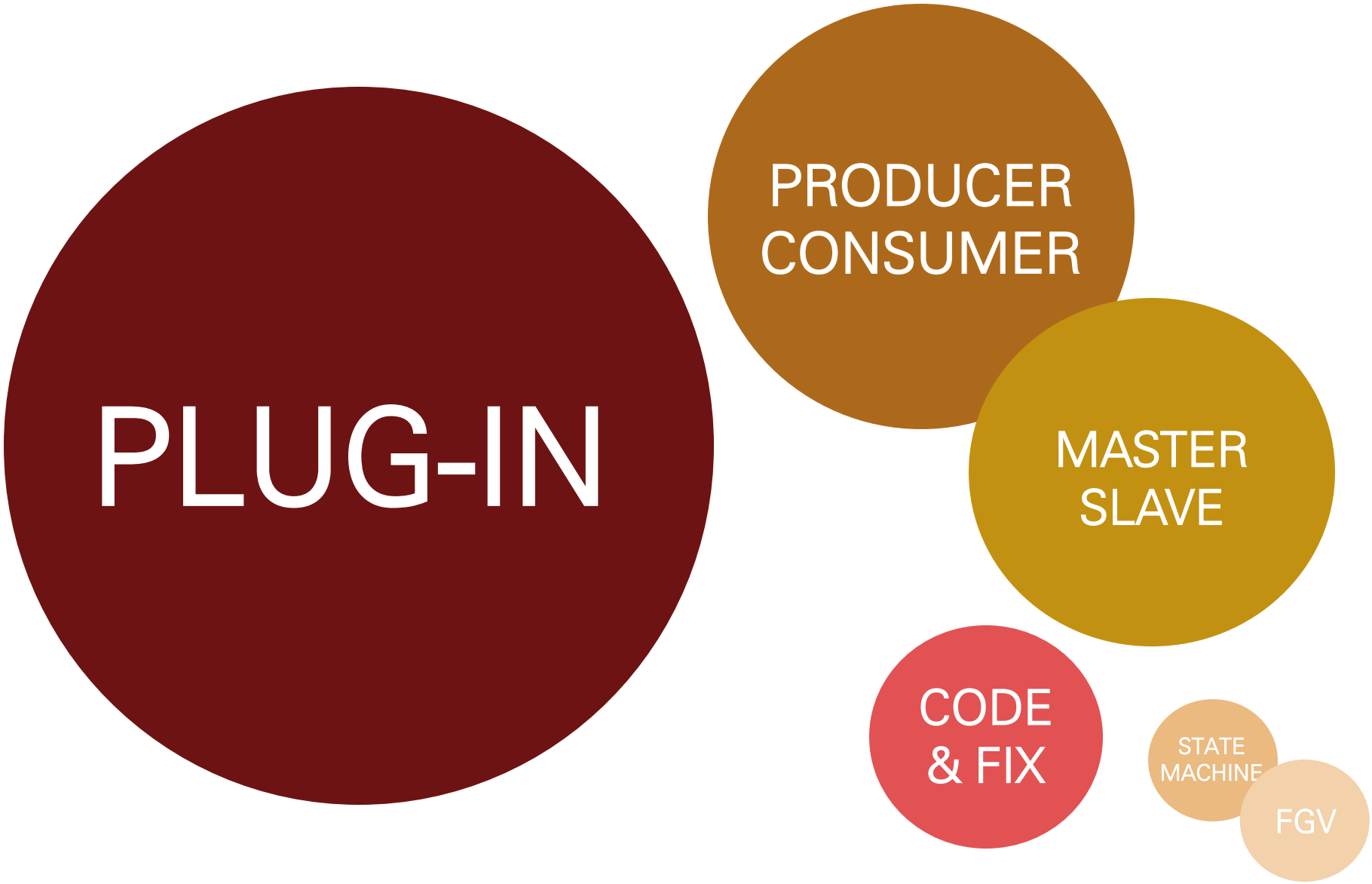


STATE MACHINE



DEMO

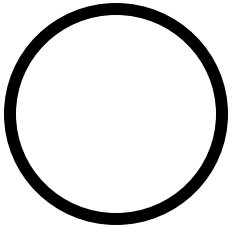
JUST USE
STATECHART!!


PLUG-IN
PRODUCER
CONSUMER
MASTER
SLAVE
CODE
& FIX
STATE
MACHINE
FGV


KBS
ARREST!!



CUT



OPEN

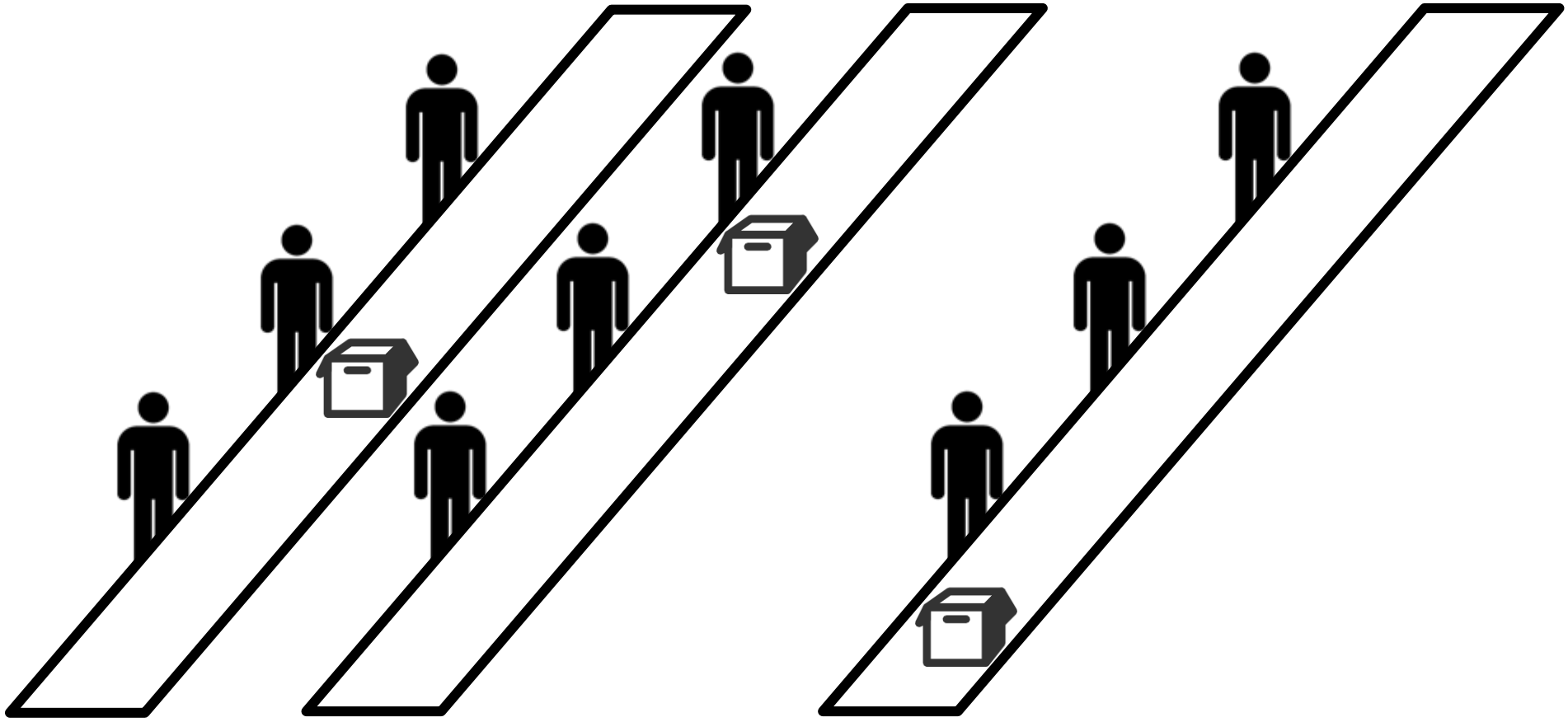


PUT

CLOSE



TEST LINE



TEST SEQUENCE



TEST DEVICE



DAQ



DMM



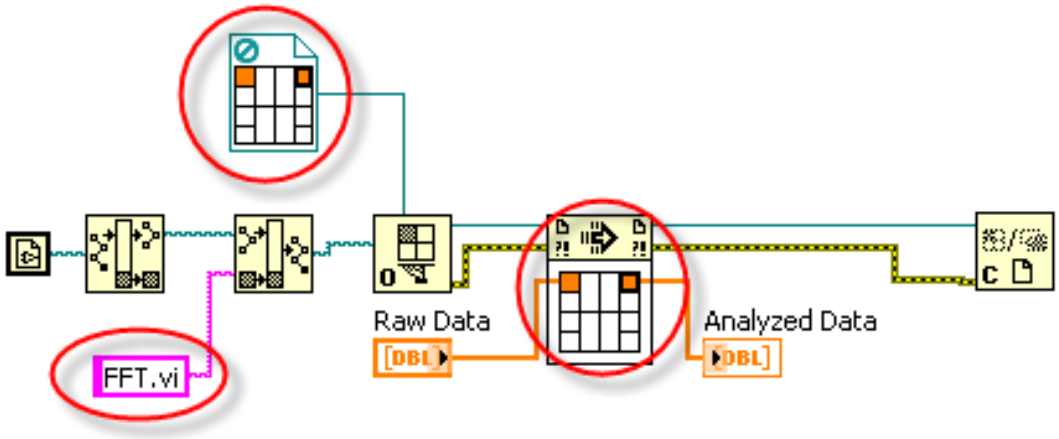
SCOPE

TEST LINE 1

TEST LINE 2

TEST LINE N

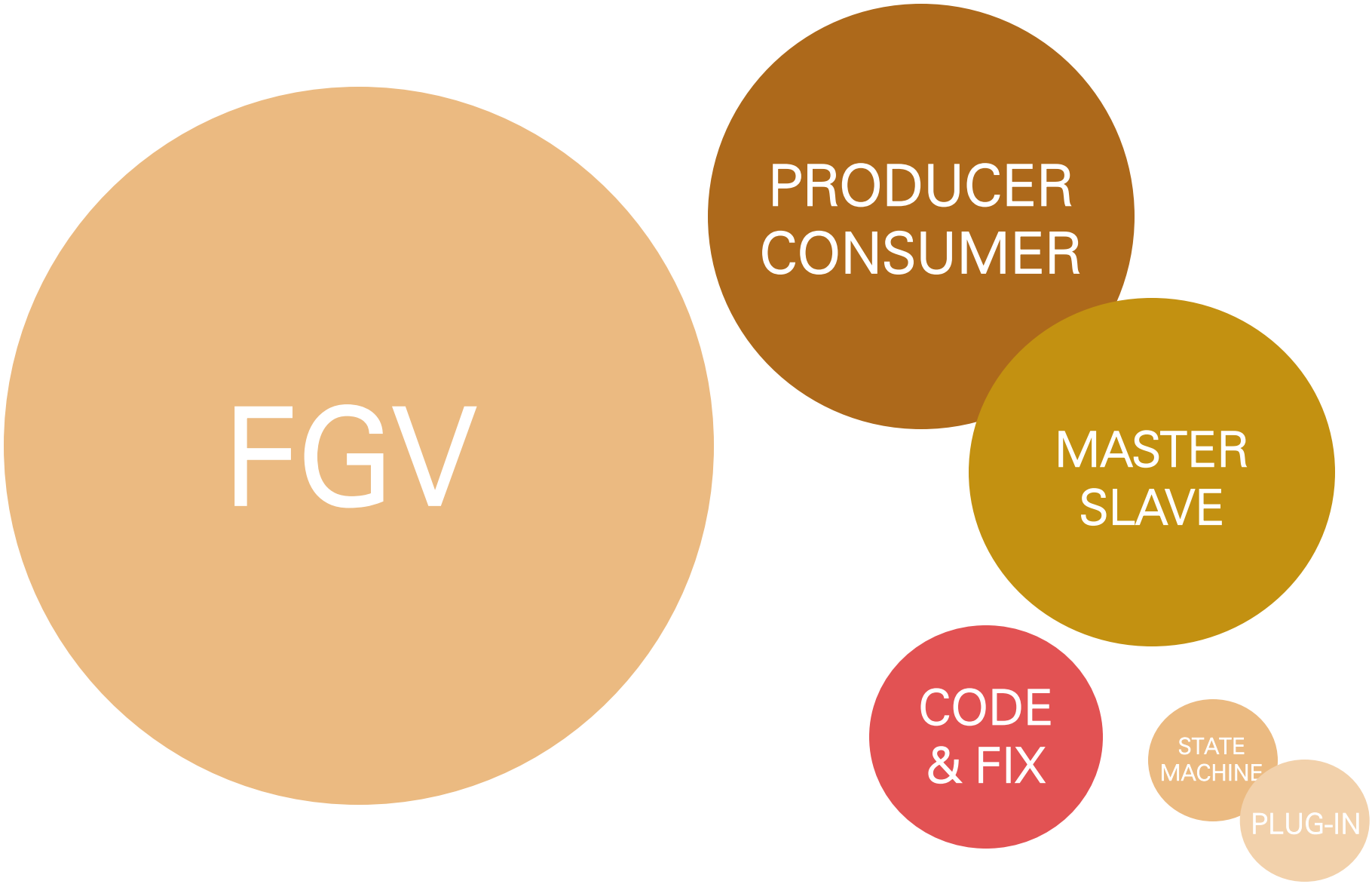
PLUG-IN ARCHITECTURE USING VI-SERVER

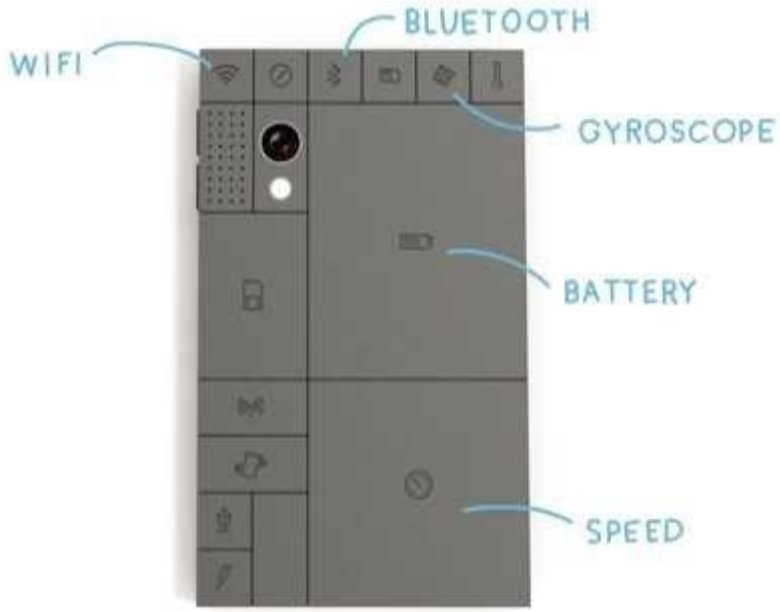


DEMO

CONTROL

VI SERVER!!


FGV
PRODUCER
CONSUMER
MASTER
SLAVE
CODE
& FIX
STATE
MACHINE
PLUG-IN

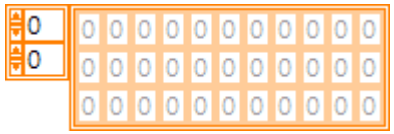


MODULE...



DATA + METHODS

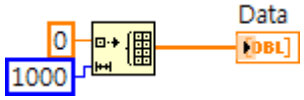
DATA



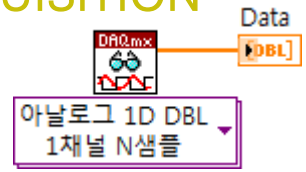
+

METHODS

INITIALIZE



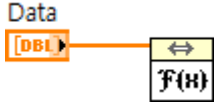
ACQUISITION



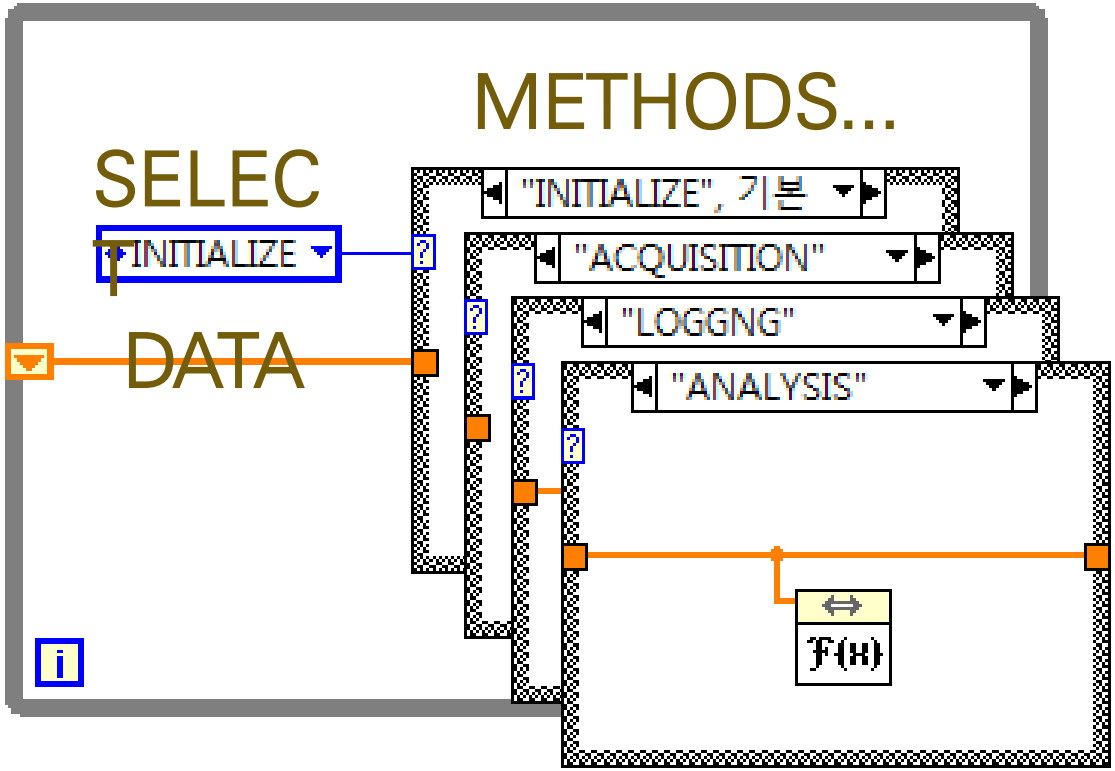
LOGGING



ANALYSIS



INCAPSULATION



DEMO

INCAPSULATE FGV

Q&A